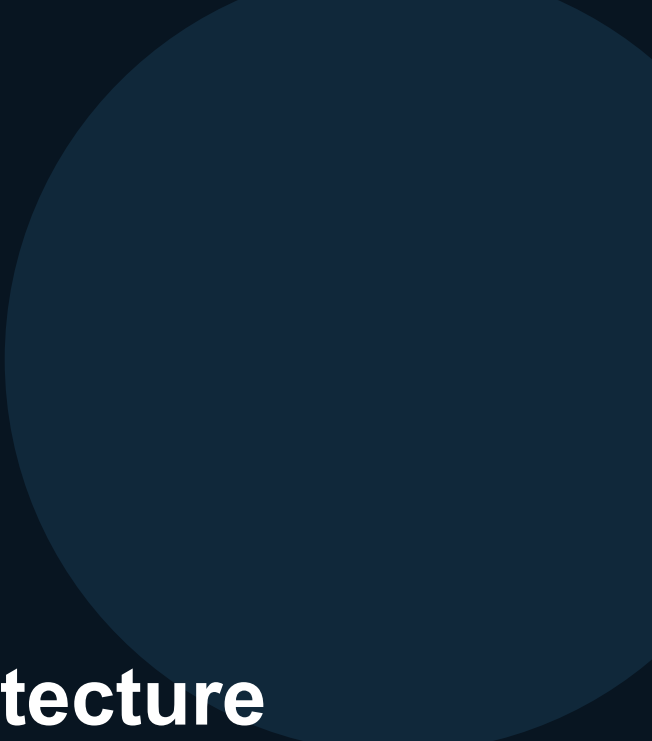




QUANT'S TRADE

Platform Architecture & Feature Specification

A complete product blueprint for an evidence-driven
quantitative research and trading cockpit

PRODUCT ARCHITECTURE / SECURITY / QUANT ENGINE / EXPERIENCE



Contents

The platform is organized by product domain rather than implementation detail.

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1. Platform Foundation

Authentication

- Email and password authentication
- Secure login and registration
- Session management
- Access and refresh tokens
- HttpOnly authentication cookies
- Secure production cookies
- SameSite cookie protection
- CSRF protection
- Protected API routes
- Role-based permissions
- Page-level permissions
- Admin-only endpoints
- User ownership validation
- Multi-user data isolation
- Demo access controls
- Account verification
- Account suspension
- Secure logout
- Authentication rate limiting

User Management

- User profile
- Account status
- User settings
- Broker preferences
- Trading preferences
- Risk preferences
- AI preferences
- Memory preferences
- Notification preferences
- Market-universe preferences
- Trading-horizon preferences
- Execution-mode preferences
- Page-access management
- Admin-mode controls
- User activity audit history

Database

- PostgreSQL
- Prisma ORM
- Versioned migrations
- Transaction support
- Relational data integrity
- User-scoped ownership
- Audit records
- Immutable ledger events
- Soft deletion where appropriate
- Strategy version history
- Execution records
- Broker order records
- Market-data snapshots
- Research history
- AI invocation history
- Memory and embedding storage

Caching

- Redis-compatible cache architecture
- In-memory cache fallback
- TTL caching
- LRU-style caching
- User-scoped caching
- Quote caching
- Price-history caching

- Market-data caching
- Cache invalidation
- Provider-aware cache keys
- In-flight request deduplication
- Data-freshness metadata
- Cache health metrics

2. Trading Command Center

Dashboard

- Portfolio value
- Available cash
- Buying power
- Daily P&L;
- Total P&L;
- Open positions
- Market exposure
- Sector exposure
- Risk status
- Current market regime
- Scanner summary
- Signal distribution
- Top opportunities
- Pending approvals
- Active alerts
- Recent trades
- Broker status
- Market-data status
- Python-engine status
- Scan-job status

Command Center Experience

- Global market status bar
- Broker connection indicator
- Data-freshness indicator
- Engine health indicator
- Paper and live mode indicators
- Quick actions
- Universal symbol search
- Stock workspace drawer
- Command palette
- Customizable dashboard
- Multi-monitor layouts
- Full-screen trading mode
- Daily market briefing
- What changed? summary
- Session-aware workflow

3. AI Platform

AI Infrastructure

- Provider abstraction
- OpenAI integration
- Central AI service
- Feature-specific AI routing
- Structured JSON outputs
- Prompt registry
- Prompt versioning
- Prompt hashing
- Schema validation
- Input hashing
- Invocation logging
- Invocation persistence
- Token accounting

- Cost estimation
- User budgets
- Feature budgets
- Concurrency limits
- Rate limiting
- Retry handling
- Retry-After support
- Circuit breaker
- Health monitoring
- Diagnostics
- Admin metrics
- Model configuration
- Graceful provider failure

AI Security

- Prompt redaction
- Secret redaction
- Broker-data redaction
- Strategy redaction
- Personally identifiable information redaction
- User-context isolation
- Feature permissions
- Input validation
- Output validation
- Response-schema enforcement
- AI rate limiting
- Audit logging
- State-changing action restrictions
- Human approval boundaries
- Prevention of direct AI trade execution

AI Features

- General trading chat
- Stock analysis
- Portfolio analysis
- Trade review
- News summaries
- Scanner-result explanations
- Strategy generation
- Strategy explanation
- Strategy comparison
- Strategy editing
- Strategy research
- Portfolio recommendations
- Portfolio scenarios
- Matrix recommendations
- Matrix scenarios
- Research reports
- Research questions
- Thesis review
- Market-impact analysis
- Strategy-impact analysis
- Portfolio-impact analysis
- Evidence-backed recommendations

4. Memory Agent

Memory Storage

- Memory events
- Memory categories
- Memory links
- Memory feedback
- Memory profiles
- User preferences
- Behaviour patterns

- Historical decisions
- Strategy memories
- Portfolio memories
- Research memories
- Trade memories
- Broker memories
- Approval memories
- Risk memories
- Scanner memories
- System memories

Vector Memory

- Vector embeddings
- OpenAI-compatible embeddings
- Background embedding pipeline
- Embedding queue
- Retry handling
- Failed-embedding recovery
- Staleness detection
- Re-embedding
- Embedding exclusions
- Vector diagnostics
- Backfill support
- Dimension validation

Hybrid Retrieval

- Vector similarity
- Full-text search
- Exact-entity matching
- Category matching
- Recency
- Relevance scoring
- Confidence scoring
- Outcome weighting
- User feedback
- Deduplication
- Provenance
- Context assembly
- Retrieval auditing

Personalization

- Explicit preferences
- Inferred preferences
- Behaviour patterns
- Confidence scoring
- Contradicting evidence
- Cold-start handling
- Staleness handling
- Memory controls
- Archive
- Exclude
- Delete
- Reset
- Rebuild
- Export
- Outcome-based validation

Copilot Integration

- Strategy Copilot
- Portfolio Copilot
- Research Copilot
- Matrix Copilot
- Scanner explanations
- Trade reviews
- Playbook recommendations

5. Strategy Lab

Strategy Builder

- Strategy name and description
- Trading universe
- Trading horizon
- Entry rules
- Exit rules
- Risk rules
- Position-sizing rules
- Validation rules
- Technical indicators
- Logical groups
- Nested conditions
- AND and OR logic
- Parameter controls
- Builder validation
- Inline error messages
- Strategy preview
- Natural-language strategy generation
- Load AI drafts into the builder

Strategy DSL

- Strategy JSON contract
- Shared frontend and backend contract
- DSL validation
- Strategy compiler
- Strategy interpreter
- Execution engine
- Condition evaluation
- Rule normalization
- Contract versioning
- Safe unsupported-rule rejection
- Reusable Python execution

Backtesting

- Single-symbol backtesting
- Multi-symbol backtesting
- Shared-capital testing
- Portfolio backtesting
- Walk-forward testing
- Parameter sweeps
- Monte Carlo simulation
- Robustness testing
- Stress testing
- Overfitting detection
- Regime-specific analysis
- Slippage
- Spread
- Fees
- Commission
- Exposure
- Turnover
- Drawdown
- CAGR
- Sharpe ratio
- Sortino ratio
- Calmar ratio
- Win rate
- Profit factor
- Expectancy
- Trade analytics
- Equity curve
- Drawdown curve
- Benchmark comparison

Strategy Lifecycle

- Draft
- Research
- Candidate
- Active
- Archived
- Deployment readiness
- Active strategy set
- Activation
- Deactivation
- Rollback
- Immutable versions
- Version comparison
- Canonical ownership
- Audit history
- Evidence history
- Performance history

Strategy Copilot

- Natural-language strategy generation
- Explain strategy
- Review strategy
- Compare versions
- Answer strategy questions
- Generate research reports
- Propose strategy edits
- Approve proposed edits
- Reject proposed edits
- Load approved changes into builder
- Evaluate deployment readiness
- Identify weaknesses
- Highlight overfitting risk
- Explain supporting evidence

6. Matrix Lab

Strategy Matrix

- Strategy-to-sector routing
- Strategy-to-regime routing
- Capital allocation percentage
- Equal weighting
- Manual weighting
- Risk-parity weighting
- Confidence weighting
- Evidence weighting
- Volatility-adjusted weighting
- Active cells
- Pending cells
- Sit-out cells
- Blocked cells
- Active strategy set
- Deployment ownership
- Allocation snapshots
- Allocation audit history

Matrix Replay

- Replay historical market data
- Resolve historical regime
- Resolve matrix routing
- Execute selected strategies
- Apply shared capital
- Measure portfolio outcome

Matrix Copilot

- Matrix review
- Cell optimization
- Coverage-gap detection
- Regime analysis
- Allocation suggestions
- Deployment reasoning
- Alternative comparison
- Scenario simulation
- Proposal creation
- Proposal naming
- Proposal duplication
- Proposal approval
- Proposal rejection
- Proposal restoration
- Historical evidence explanation

7. Portfolio Lab

Portfolio

- Holdings
- Positions
- Cash
- Buying power
- Allocation
- Sector exposure
- Concentration
- Performance
- Realized P&L;
- Unrealized P&L;
- Drawdown
- Risk metrics
- Position sizing
- Trade history
- Paper portfolio
- Broker portfolio
- Portfolio reconciliation

Portfolio Simulation

- Multiple strategies
- Shared capital
- Allocation weighting
- Correlation
- Concentration
- Position overlap
- Drawdown
- Equity curves
- Proposed-trade impact
- Portfolio scenarios
- Before-and-after comparisons
- Estimated transaction costs
- Risk impact
- Cash impact

Portfolio Copilot

- Portfolio review
- Allocation recommendations
- Diversification analysis
- Concentration analysis
- Risk analysis
- Scenario analysis
- Proposed portfolio changes
- Proposal comparison
- Proposal storage
- Proposal approval

- Proposal rejection
- Evidence explanation
- Natural-language portfolio questions

8. Risk Engine

Portfolio Risk

- Position exposure
- Sector exposure
- Strategy exposure
- Market exposure
- Concentration risk
- Correlation risk
- Volatility risk
- Drawdown risk
- Cash utilization
- Portfolio beta
- Value at Risk
- Expected shortfall
- Stress scenarios
- Risk contribution
- Liquidity risk
- Maximum position size
- Maximum sector exposure
- Maximum daily loss
- Maximum portfolio drawdown

Trade Risk

- Position-aware analysis
- Existing exposure checks
- Duplicate-position checks
- Position-sizing validation
- Stop-loss validation
- Reward-to-risk validation
- Liquidity validation
- Price-collar validation
- Maximum order-size validation
- Buying-power validation
- Restricted-symbol validation
- Restricted-session validation
- Fat-finger protection

Safety Controls

- Read-only mode
- Paper-broker mode
- Live dry-run mode
- Live supervised mode
- Live locked mode
- Global kill switch
- Strategy kill switch
- Broker kill switch
- Order throttling
- Duplicate-order prevention
- Broker-disconnect protection
- Automatic strategy suspension
- Daily-loss shutdown
- Drawdown shutdown
- Manual recovery workflow

9. Scanner

Scanner Capabilities

- Universe scanning
- Manual scans
- Scheduled scans
- Scan progress
- Scan cancellation
- Scan timeout
- Scan failure recovery
- Technical indicators
- Market-regime detection
- Macro-regime analysis
- AI news reasoning
- News filtering
- Risk adjustment
- Position-aware scoring
- Opportunity ranking
- BUY signals
- HOLD signals
- SELL signals
- Confidence scores
- Opportunity scores
- Signal explanations
- Filter chips
- Saved views
- Stock-detail drawer
- Price charts

Stock Universes

- Manual universe
- Watchlist universe
- Sector universe
- Industry universe
- S&P; 500 sample
- Custom screen
- User-owned universes
- Universe members
- Universe activation
- Universe editing

Future Scanner Modules

- Premarket scanner
- Intraday scanner
- Closing scanner
- After-hours scanner
- Gap scanner
- Breakout scanner
- Momentum scanner
- Mean-reversion scanner
- Relative-strength scanner
- Unusual-volume scanner
- Volatility scanner
- Market-breadth scanner
- Sector-rotation scanner
- Earnings-event scanner
- Options-liquidity scanner

10. Market Data Platform

Providers

- Yahoo market data
- Moomoo market data
- Scanner-data fallback

- Provider abstraction
- Provider health monitoring
- Provider capability detection
- Automatic fallback
- Provider-specific diagnostics

Market Data

- Live quotes
- Delayed quotes
- Historical prices
- Chart data
- Market metadata
- Exchange metadata
- Price-history normalization
- Data persistence
- Market-data snapshots
- Data-source headers
- Data-freshness timestamps
- Cache status
- Provider status

Future Data Capabilities

- Real-time streaming
- WebSocket delivery
- Corporate actions
- Dividends
- Stock splits
- Earnings calendars
- Economic calendars
- Point-in-time fundamentals
- Options chains
- Futures data
- FX data
- Cryptocurrency data
- Data-quality anomaly detection
- Provider entitlement management

11. Research Lab

Research Workspace

- Research projects
- Project status
- Research reports
- AI-generated reports
- Research notes
- Research questions
- Investment theses
- Thesis review
- Version comparison
- Research history
- Source tracking
- Evidence tracking
- Project archiving
- Project restoration
- Symbol research
- Sector research
- Theme research
- Company research
- Market research

Research Copilot

- Research overview
- Ask research questions
- Symbol analysis
- Sector analysis

- Scanner summary
- Watchlist summary
- Upcoming-event analysis
- Deep research
- Market-impact analysis
- Portfolio-impact analysis
- Strategy-impact analysis
- Matrix-impact analysis
- Scanner-impact analysis
- Company comparison
- Theme analysis
- Thesis challenge
- Evidence explanation

12. Trading and Execution

Internal Paper Trading

- Proposed trades
- Paper orders
- Paper positions
- Paper trades
- Paper fills
- Transaction ledger
- Fees
- Cash deposits
- Cash withdrawals
- Mark-to-market events
- Position closure
- Realized P&L;
- Unrealized P&L;
- Portfolio updates
- Ledger rebuilding
- Ledger integrity verification

Pre-Trade Workflow

- Opportunity
- Pre-trade analysis
- Proposed trade
- Risk validation
- Approval request
- Paper execution
- Ledger update
- Outcome review

Approval System

- Manual approval
- Semi-automated approval
- Full-automation configuration
- Pending status
- Approved status
- Rejected status
- Snoozed status
- Executed status
- Approval notes
- Approval history
- Approval audit events
- Required approval before execution
- User ownership enforcement

13. Broker Integration

Broker Architecture

- Broker-adapter contract
- Broker registry
- Broker capability detection
- Broker configuration service
- Encrypted broker secrets
- Connection diagnostics
- Broker health checks
- Broker connection logs
- Account synchronization
- Position synchronization
- Order synchronization
- Fill synchronization
- Ledger reconciliation
- Broker execution audits

Current Integrations

- Moomoo adapter architecture
- IBKR connection diagnostics
- IBKR configuration
- Broker paper execution
- Order preview
- Paper-order placement
- Order status tracking
- Order cancellation
- Order modification
- Fill retrieval
- Account retrieval
- Reconciliation

Execution Modes

- Read only
- Internal paper
- Broker paper
- Live dry run
- Live supervised
- Live locked

Live Execution Note

- IBKR should remain connection-diagnostic or paper-only until the complete live-execution path has been independently tested and certified.

14. Alerts and Notifications

Alert System

- Scanner alerts
- Approval alerts
- Risk alerts
- Portfolio alerts
- Engine alerts
- Broker alerts
- Validation alerts
- System alerts
- Severity levels
- Alert lifecycle
- Active alerts
- Acknowledged alerts
- Snoozed alerts
- Resolved alerts
- Expired alerts
- Alert rules
- Alert delivery history

- Delivery retries
- Alert health monitoring

Current Notification Channels

- In-app notifications
- Telegram notifications
- Telegram verification
- Telegram test delivery

Future Notification Channels

- Email
- SMS
- Mobile push
- Slack
- Microsoft Teams
- Webhooks
- Daily digests
- Weekly digests
- Quiet hours
- Escalation policies
- Duplicate suppression

15. Validation and Evidence

Signal Validation

- Validation signals
- Pending evaluations
- Matured evaluations
- Failed evaluations
- Skipped evaluations
- Confidence calibration
- Regime validation
- Sector validation
- Historical signal outcomes
- Evidence aggregation
- Performance by signal type
- Performance by market regime
- Performance by sector
- Evidence-driven deployment readiness

Trust and Reproducibility

- Backtest evidence
- Strategy evidence
- Copilot evidence
- Research evidence
- Portfolio evidence
- Matrix evidence
- Source provenance
- Data-source metadata
- Historical context disclosure
- Run timestamps
- Strategy version tracking
- Configuration tracking
- Audit history

16. Playbook

Playbook Management

- Multiple playbooks
- Versioned playbooks
- Playbook versions
- Playbook snapshots
- Playbook restoration
- Playbook synchronization

- Playbook comparison
- Version promotion
- Playbook exports
- Export history
- Playbook run history

Playbook Intelligence

- Evidence rebuilding
- Playbook insights
- AI recommendations
- Recommendation review
- Accept recommendation
- Reject recommendation
- Convert recommendation to draft
- Outcome-driven learning
- Strategy lessons
- Portfolio lessons
- Risk lessons
- Scanner lessons
- Trade-review lessons

17. Analytics

Strategy Analytics

- Returns
- Drawdown
- Sharpe ratio
- Sortino ratio
- Win rate
- Profit factor
- Expectancy
- Trade duration
- Exposure
- Turnover
- Regime performance
- Sector performance
- Parameter sensitivity
- Robustness
- Overfitting risk

Portfolio Analytics

- Equity curve
- Portfolio drawdown
- Allocation
- Exposure
- Concentration
- Correlation
- Sector attribution
- Strategy attribution
- Risk contribution
- Cash utilization
- Realized P&L;
- Unrealized P&L;

Execution Analytics

- Orders
- Fills
- Rejections
- Cancellation rate
- Slippage
- Spread cost
- Commission
- Execution latency
- Broker errors
- Reconciliation differences

18. Security

Application Security

- Authentication
- Authorization
- Role-based access
- Page-level access
- User isolation
- Ownership validation
- CSRF protection
- Credentialed CORS
- Origin restrictions
- Content Security Policy
- Helmet security headers
- HSTS
- Secure cookies
- HttpOnly cookies
- SameSite cookies
- Request-size limits
- Rate limiting
- Request IDs
- Input validation
- Output validation
- Error sanitization
- Secret redaction
- Sensitive-output redaction
- Runtime safety checks

Data Security

- Encrypted broker secrets
- Encryption-key rotation
- Encrypted strategy storage
- User-scoped records
- Audit records
- No-store API responses
- Secure environment variables
- No hardcoded credentials
- Sensitive-data classification
- Backup encryption
- Data-retention controls
- Account-data deletion
- Memory exclusion controls

Production Security

- Multi-factor authentication
- Passkeys
- Trusted-device management
- Session management
- Login anomaly detection
- Refresh-token reuse detection
- Account lockout
- Cloud secret manager
- Cloud key-management service
- Tamper-evident audit logs
- Dependency scanning
- Secret scanning
- Static security analysis
- Dynamic security testing
- Penetration testing
- Threat modelling
- Incident-response runbooks
- Key-rotation runbooks
- Broker IP allowlists
- Separate development, paper and live environments

19. Observability

Platform Monitoring

- Runtime diagnostics
- Engine diagnostics
- Broker diagnostics
- Market-data diagnostics
- Security diagnostics
- AI diagnostics
- Memory diagnostics
- Health endpoints
- Readiness endpoints
- Liveness endpoints
- Structured logs
- Request IDs
- Error tracking
- Performance metrics
- Distributed tracing
- Service dashboards
- Incident alerts

AI Monitoring

- AI invocation count
- AI failures
- AI latency
- AI cost
- Token usage
- Rate-limit events
- Circuit-breaker status
- Model usage
- Feature usage
- Invalid output rate
- Prompt version
- Schema version

Operational Monitoring

- Scan-job metrics
- Queue metrics
- Cache metrics
- Market-data freshness
- Provider availability
- Broker availability
- Order status
- Execution failures
- Ledger reconciliation
- Embedding metrics
- Retrieval metrics
- Alert-delivery metrics
- Database health

20. Architecture

Frontend

- React
- Vite
- Lazy-loaded pages
- Component-based architecture
- Feature-based organization
- Shared API client
- Fetch-based requests
- Request deduplication
- Polling coordination
- Responsive layouts
- Reusable chart components

- Reusable data cards
- Design-token system
- Dark and light themes
- Premium trading theme
- Accessibility support

Backend

- Node.js
- Express
- REST APIs
- Feature-based modules
- Controllers
- Services
- Repositories
- Middleware
- Prisma data layer
- Central authentication
- Central authorization
- Central AI platform
- Central memory platform
- Broker-adapter layer
- Market-data hub
- Execution guards
- Background jobs
- Audit services

Python Engine

- Scanner
- Indicators
- Strategy execution
- Strategy interpretation
- Backtesting
- Portfolio backtesting
- Walk-forward testing
- Matrix replay
- Regime detection
- Macro-regime analysis
- Risk calculations
- Position-aware analysis
- Pre-trade analysis
- Paper execution
- Broker bridges
- News filtering
- Overfitting detection
- Safety management
- Runtime artifacts

Infrastructure

- PostgreSQL
- Prisma ORM
- Vector storage
- Redis-compatible cache
- Background workers
- Object storage for artifacts
- Central secrets management
- Central logging
- Metrics
- Tracing
- Automated backups
- Point-in-time recovery
- Continuous integration
- Continuous deployment
- Feature flags

Platform Flow

- Frontend
- Express API
- Authentication and authorization
- Application services
- Python quantitative engine
- Market data and broker adapters
- PostgreSQL, cache and audit storage
- Alerts, analytics and AI copilots

21. Visual Design and User Experience

Visual System

- Premium dark trading interface
- Professional light theme
- Consistent color system
- Consistent spacing
- Consistent typography
- Consistent border radius
- Consistent elevation
- Restrained profit and loss colors
- High-density information layouts
- Clear information hierarchy
- Responsive design
- Accessible contrast
- Reduced-motion support

Charts and Visualization

- Candlestick charts
- Volume charts
- Indicator overlays
- Entry and exit markers
- Equity curves
- Drawdown charts
- Exposure charts
- Allocation charts
- Correlation heatmaps
- Strategy-regime heatmaps
- Sector treemaps
- P&L; attribution
- Portfolio risk radar
- Signal-distribution charts
- Opportunity-ranking charts
- Interactive tooltips
- Zoom and time-range controls

Interaction Design

- Skeleton loading states
- Empty states
- Error recovery
- Optimistic updates
- Confirmation dialogs
- Destructive-action warnings
- Smooth page transitions
- Purposeful micro-animations
- Scan-progress animations
- Keyboard navigation
- Command palette
- Universal search
- Contextual quick actions
- Saved workspace layouts
- One-click report exports
- Mobile summary mode

- Desktop research mode
- Multi-monitor trading mode

22. Reliability and Operations

Capabilities

- Background job queue
- Job retries
- Job timeouts
- Dead-letter queue
- Graceful shutdown
- Job recovery
- Idempotent scan requests
- Idempotent approval requests
- Idempotent execution requests
- Provider failover
- Broker-disconnect recovery
- Database migration procedures
- Database rollback procedures
- Automated backups
- Disaster recovery
- Feature flags
- Paper and live isolation
- Service-level objectives
- Incident timeline
- Platform status page

23. Testing and Governance

Testing

- Python unit tests
- Node service tests
- React component tests
- API contract tests
- Strategy regression tests
- Golden market datasets
- Ledger-integrity tests
- Ownership-isolation tests
- Authentication tests
- Security-header tests
- Broker sandbox tests
- Paper-execution tests
- End-to-end workflow tests
- Responsive-layout tests
- Accessibility tests
- Visual regression tests
- Load tests
- Provider-failure tests

Model and Strategy Governance

- Strategy approval workflow
- Strategy deployment evidence
- Strategy version history
- Model version tracking
- Prompt version tracking
- AI evaluation datasets
- Backtest reproducibility
- Data-source provenance
- Configuration hashes
- Strategy rollback
- Human execution approval
- Independent live-trading certification
- Periodic strategy review
- Automatic evidence-decay warnings

24. Major Platform Modules

Complete Module Map

- Authentication
- User Management
- Trading Command Center
- AI Platform
- Memory Agent
- Strategy Lab
- Matrix Lab
- Portfolio Lab
- Risk Engine
- Scanner
- Market Data Platform
- Research Lab
- Internal Paper Trading
- Broker Integration
- Pre-Trade Approvals
- Alerts and Notifications
- Validation and Evidence
- Trading Playbook
- Analytics
- Security
- Observability
- Admin Diagnostics
- Design System
- Reliability
- Testing and Governance

Overall Platform Positioning

An evidence-driven quantitative trading cockpit that makes every signal, strategy, allocation, risk decision, approval and execution explainable, reviewable and auditable.

The platform is already structurally advanced for a personal or prosumer quantitative trading system. The largest remaining improvements are:

- Building a longer real-world paper-trading history
- Validating strategy performance across different regimes
- Completing production-grade broker certification
- Strengthening operational monitoring
- Adding realistic execution-cost modelling
- Improving data quality and provenance
- Polishing the visual experience
- Making the entire workflow feel like one connected trading cockpit

The Primary Operating Workflow

